Compute 2

Hannah Srzich

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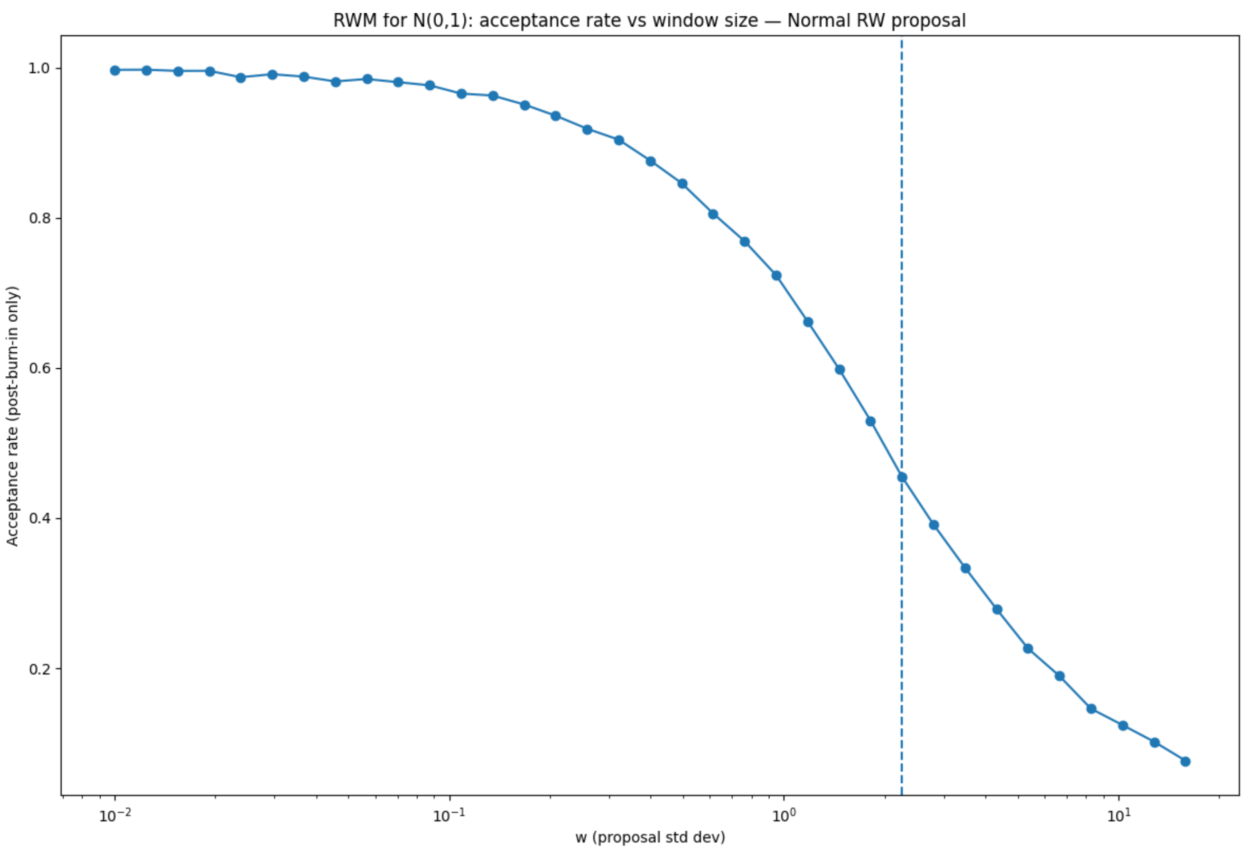
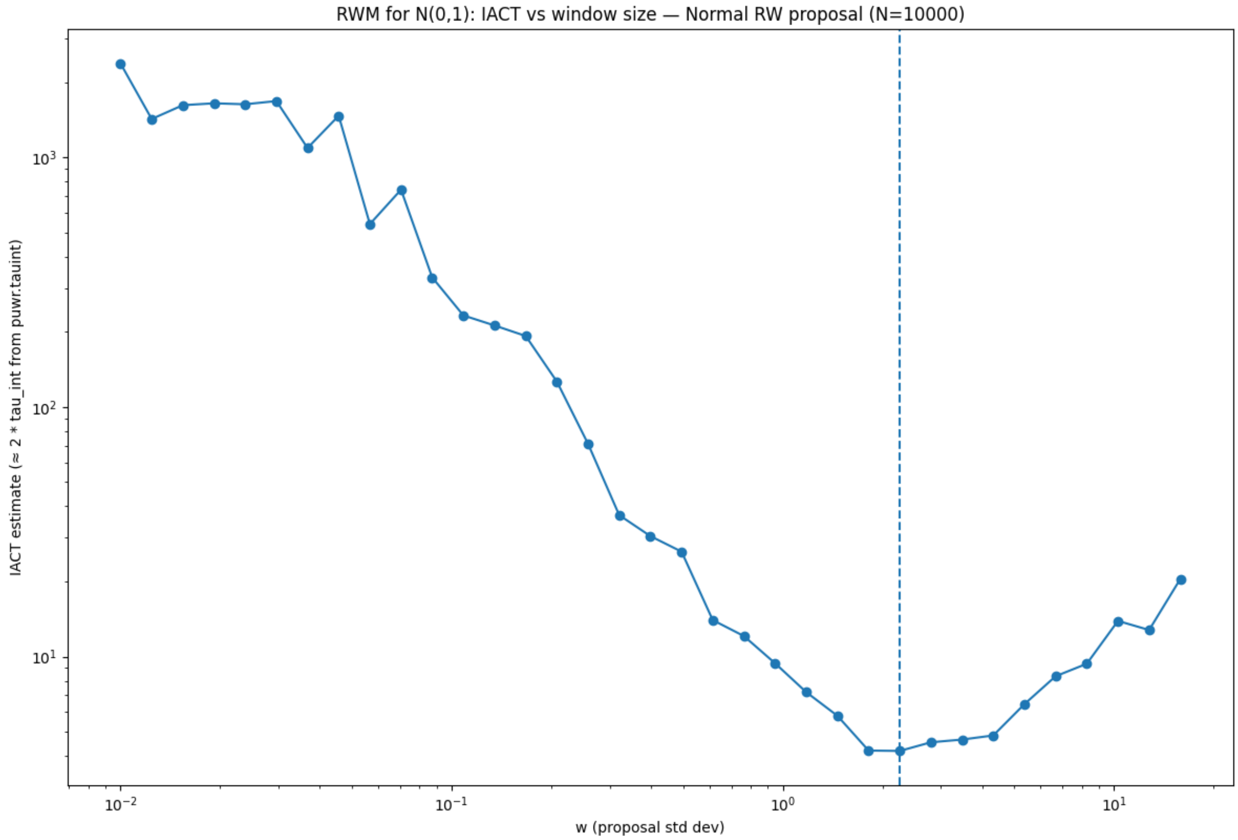
Question 1.

Computing the Integrated Auto-Correlation Time (IACT), and determining the optimal proposal window, to formalise the notion of 'good mixing'.

Consider sampling from the univariate standard normal, $N(0, 1)$. In each case generate 10,000 samples, and compute IACT for the state variable.

(a) Using the code `mcgauss` (link on paper website), or otherwise, plot IACT versus window size when using random walk Metropolis with a normal proposal. What is the optimal IACT and window size?






```
------------- Results -------------
Proposal: normal
Kept draws per run: 10,000  |  burn-in: 1,000
Best (min) IACT: 4.187
Best window size w: 2.254
Acceptance rate at best w: 0.455
ESS at best w: 2388.1
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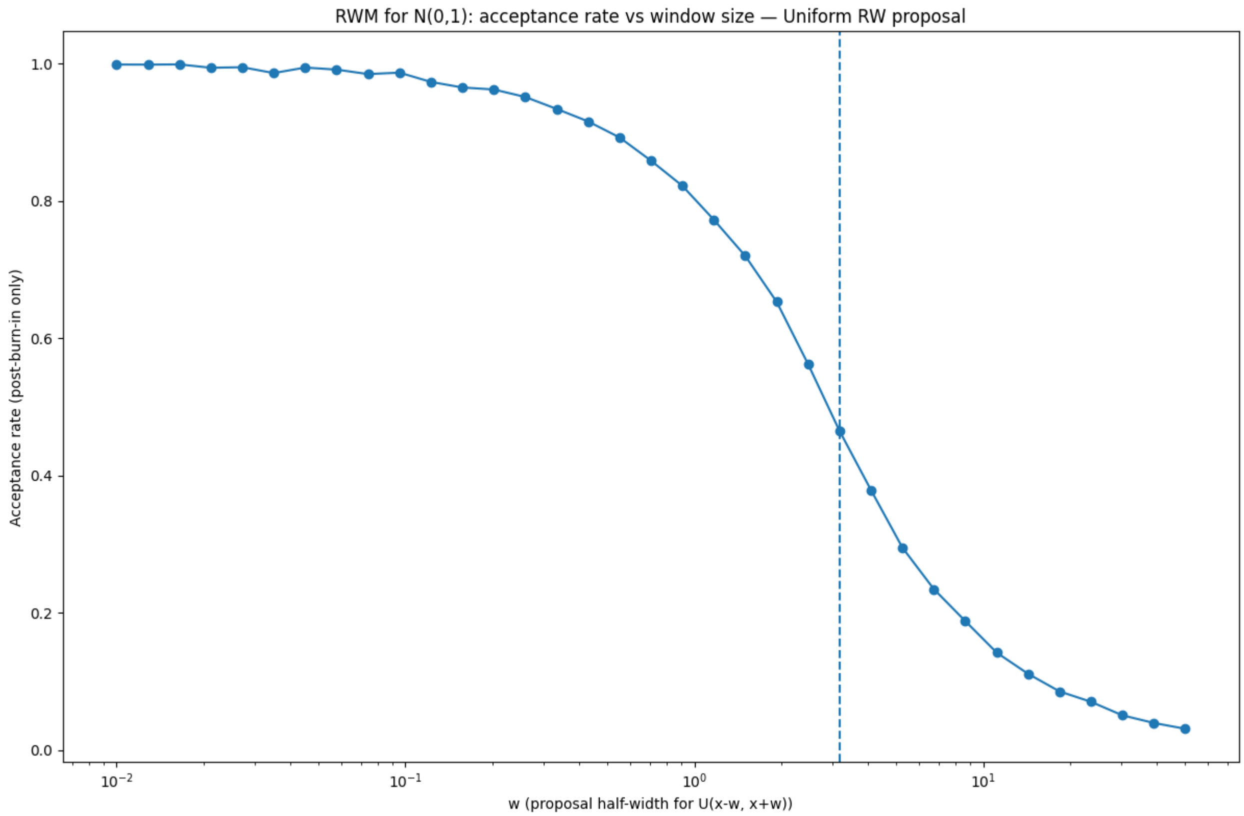
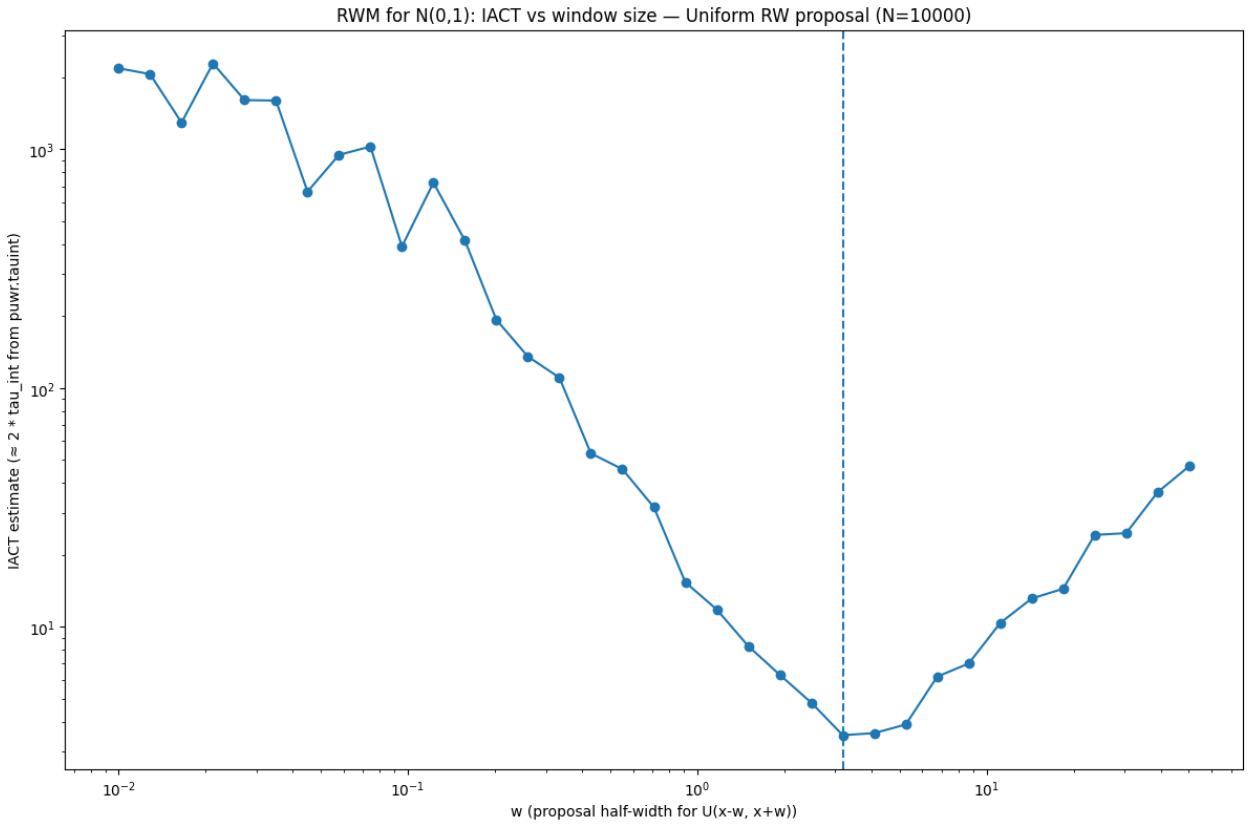
(b) Repeat the previous question, but with a uniform proposal. (Isn't that interesting).


RWM for N(0,1): acceptance rate vs window size — Uniform RW proposal
Acceptance rate (post-burn-in only)
1.0
0.8
0.6
0.4
0.2
0.0
10^{-2}
10^{-1}
10^{0}
10^{1}
w (proposal half-width for U(x-w, x+w))




```
------------- Results -------------
Proposal: uniform
Kept draws per run: 10,000  |  burn-in: 1,000
Best (min) IACT: 3.515
Best window size w: 3.184
Acceptance rate at best w: 0.465
ESS at best w: 2844.9
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```

Interesting that the IACT of the uniform proposal is better than the normal proposal and that acceptance rates are approximately the same?

(c) Download Marko Laine's `mcmcrun.m` code (see `https://mjlaine.github.io/mcmcstat/`) and use the adaptive Metropolis option (see sample code on the paper website) to sample from $N(0, 1)$. Does it achieve the optimum you found in the first part?

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------------- Part 1(c) - Adaptive Metropolis (AM) -------------
Kept draws per run: 10,000  |  burn-in: 1,000
Adaptation interval: 100
Runs: 10

IACT: mean=4.469, median=4.478, sd=0.180
Acceptance rate (kept segment): mean=0.441, median=0.440, sd=0.006
```

```
Sample mean (kept segment): mean=-0.003, median=-0.001, sd=0.013
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```

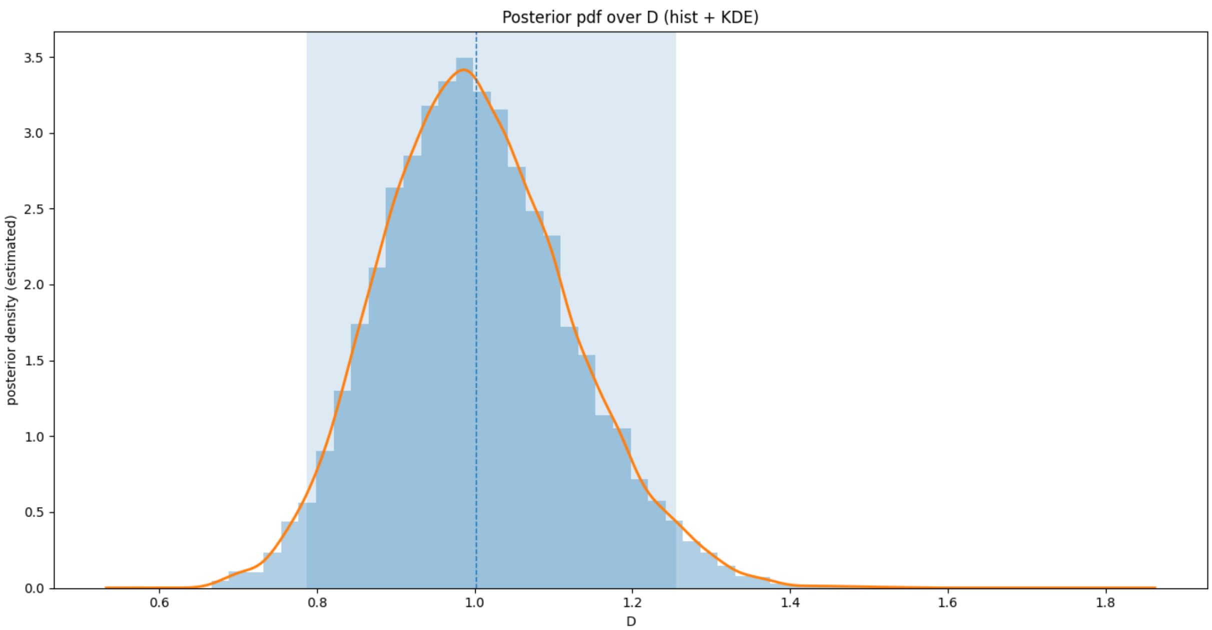
Results seem to be pretty close to the optimal IACT of 4.187, and acceptance rate of 0.455 from part (a).

Question 2.

Inverse problem for the diffusion coefficient.

MatLab code to carry out inference by MCMC for the diffusion coefficient D (an unknown constant) can be found on the paper's web page `https://coursesupport.physics.otago.ac.nz/wiki/pmwiki.php/ELEC446`.

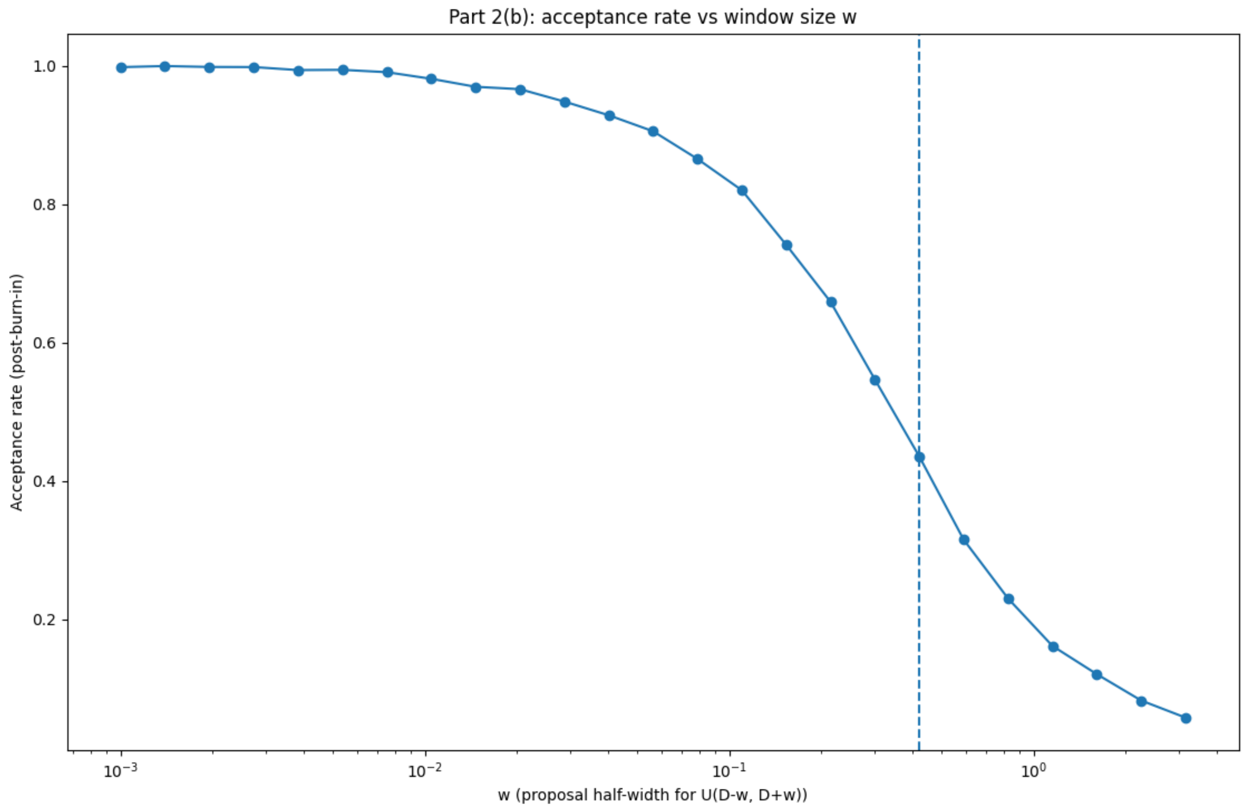
(a) Run that code (or Python equivalent) and produce a plot of the posterior pdf over D conditioned on the (simulated) measured data. Give a best value, and measure of uncertainty.

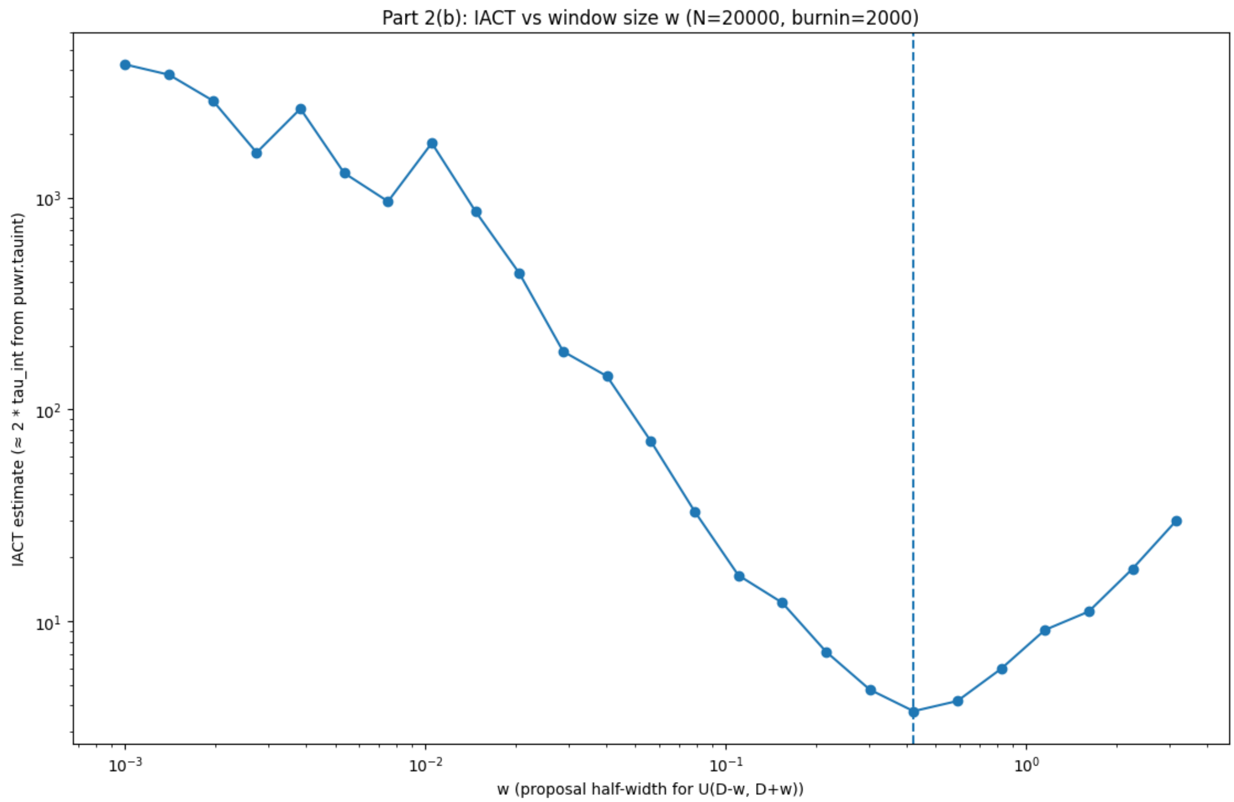



```
MCMC configuration:
  True D:        1.0
  Chain length:  20000
  Window size w:0.1
  Burn-in:       2000

Part 2(a): posterior over D given simulated data
Best value (posterior mean): 1.00218
95% credible interval: [0.787266, 1.2551]
```

(b) Plot the integrated autocorrelation time versus proposal window, and acceptance rate versus proposal window for this MCMC (for fixed simulated data). What is the optimal IACT, window size, acceptance rate?


Part 2(b): acceptance rate vs window size w
Acceptance rate (post-burn-in)
w (proposal half-width for U(D-w, D+w))
1.0
0.8
0.6
0.4
0.2
10^{-3}
10^{-2}
10^{-1}
10^{0}

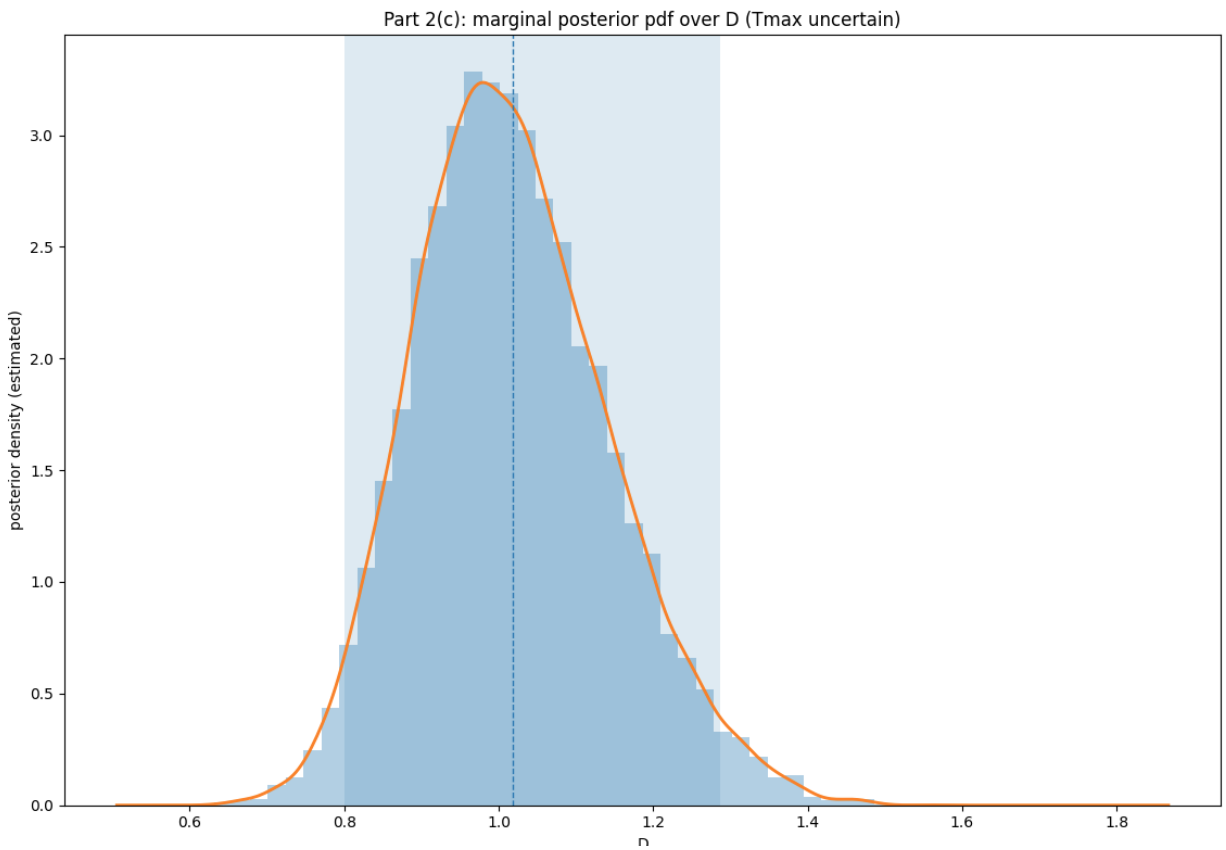



```
MCMC configuration:
  True D:        1.0
  Chain length:  20000
  Burn-in:       2000

Part 2(b): IACT vs window size w (fixed simulated data)
Best (min) IACT: 3.76132
Best window size w: 0.421697
Acceptance rate at best w (post-burn-in): 0.435246
```

(c) Using whichever you think is the best sampling algorithm available to you, modify the formulation and code to allow the final time $T_{\max}$ to be uncertain, uniformly distributed in the range [1.9, 2.1]. In particular, perform joint inference over D and the (nuisance parameter) $T_{\max}$, and plot the posterior pdf over D conditioned on the (simulated) measured data. Give a best value, and measure of uncertainty. How does the uncertainty compare to the result when $T_{\max}$ was treated as certain?

First I ran a standard Metropolis-within-Gibbs sampler...

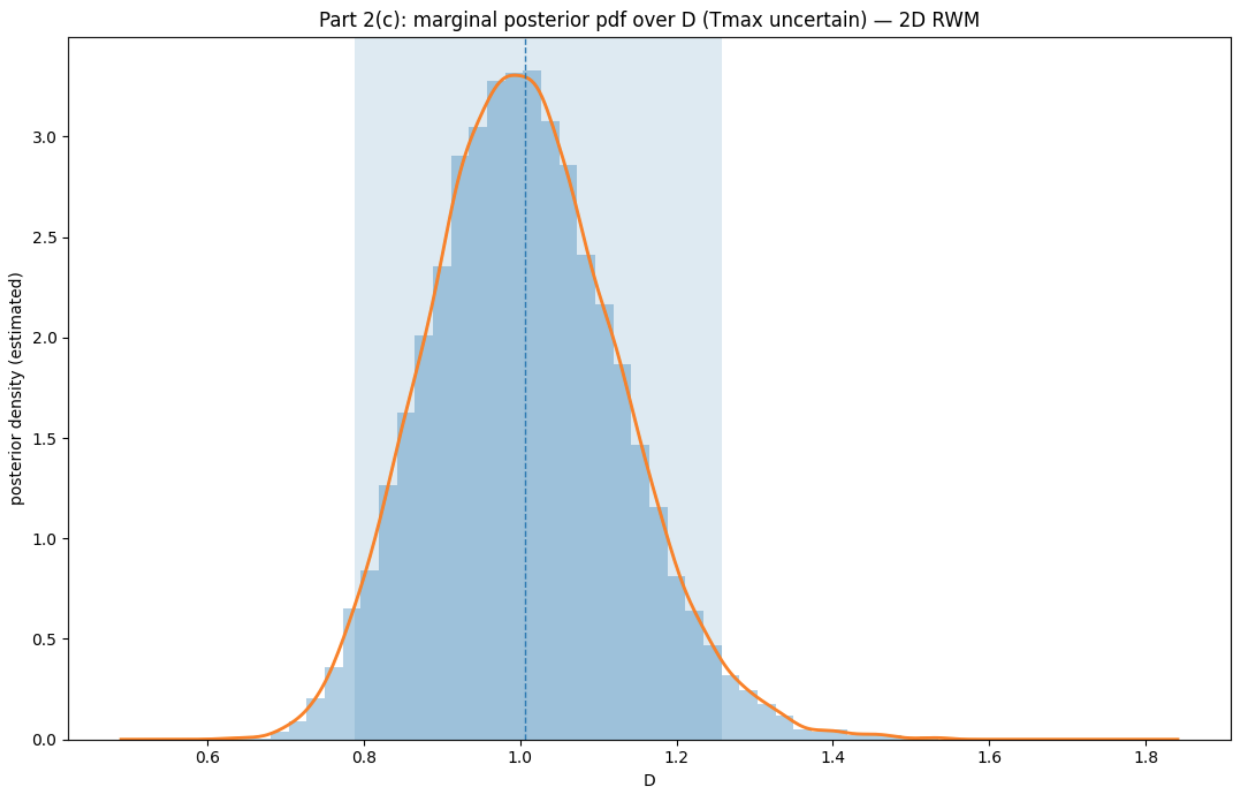



```
Part 2(c): Metropolis-within-Gibbs for joint (D, Tmax)
Configuration:
  True D:        1.0
  True Tmax:     2.0
  Chain length:  20000
  Burn-in:       2000
  wD (D RW):     0.1
  wT (T RW):     0.01
  Prior support: D>0, Tmax in [1.9, 2.1]

Posterior summaries (post burn-in):
  D mean:  1.0181   |  95% CI: [0.79979, 1.28717]
  T mean:  1.98542   |  95% CI: [1.90375, 2.09277]

Acceptance rates:
  D update (all iters):      0.8338
  Tmax update (all iters):   0.9645
  D update (post burn-in):   0.8347
  Tmax update (post burn-in):0.9646
```

Then, I ran a 2D random-walk Metropolis sampler...




```
Part 2(c): 2D Random-Walk Metropolis for joint (D, Tmax)
Configuration:
  True D:         1.0
  True Tmax:      2.0
  Chain length:   20000
  Burn-in:        2000
  wD (D RW):      0.1
  wT (T RW):      0.01
  Prior support: D>0, Tmax in [1.9, 2.1]

Posterior summaries (post burn-in):
  D mean:  1.00645   |  95% CI: [0.788214, 1.25729]
  T mean:  1.99799   |  95% CI: [1.90467, 2.09312]

Acceptance rates:
  Joint accept rate (all iters): 0.8163
  Joint accept rate (post burn-in): 0.8154
```

For the diffusion coefficient D, the fixed-$T_{\max}$ case (Part 2(a)) gives a 95% credible-interval width of 0.467834 with a best–true difference of +0.00218, the joint Metropolis-within-Gibbs case gives a width of 0.48738 with a best–true difference of +0.01810, and the joint 2D random-walk Metropolis case gives a width of 0.469076 with a best–true difference of +0.00645. It is not clear to me that there is a real difference in the uncertainty over D between the cases.

Question 3.

Some probability algebra exercises.

Consider the joint pdf $\pi(x, y, \theta)$ over x, y, and θ. Using Bayes' rule, product factoring, etc, establish the following identities between the pdfs over various conditional and marginal distributions:

(a) $\pi(\theta \mid y) = \dfrac{\pi(x, \theta \mid y)}{\pi(x \mid \theta, y)}$ (This is useful because the RHS may be evaluated at any x for which the denominator is non-zero.)

By the product factorisation of the conditional joint density $\pi(x, \theta \mid y)$ we have

$$\pi(x, \theta \mid y) = \pi(x \mid \theta, y)\, \pi(\theta \mid y).$$

Assuming that $\pi(x \mid \theta, y) > 0$, we may rearrange this to obtain that

$$\pi(\theta \mid y) = \frac{\pi(x, \theta \mid y)}{\pi(x \mid \theta, y)},$$

as required.

(b) $\pi(x, \theta \mid y) = \pi(x \mid \theta, y)\, \pi(\theta \mid y)$, i.e., we can condition the usual product factoring.

Recall the definition of a conditional joint density

$$\pi(x, \theta \mid y) = \frac{\pi(x, \theta, y)}{\pi(y)}. \tag{1}$$

We may factor the joint distribution $\pi(x, \theta, y)$ using the product rule as

$$\pi(x, \theta, y) = \pi(x \mid \theta, y)\, \pi(\theta, y). \tag{2}$$

Substituting (2) into (1) gives

$$\pi(x, \theta \mid y) = \frac{\pi(x \mid \theta, y)\, \pi(\theta, y)}{\pi(y)}. \tag{3}$$

But by definition,

$$\pi(\theta \mid y) = \frac{\pi(\theta, y)}{\pi(y)}.$$

So, from (3) we have

$$\pi(x, \theta \mid y) = \pi(x \mid \theta, y)\, \pi(\theta \mid y),$$

as required.

(c) $\pi(\theta \mid y) = \dfrac{\pi(x, y \mid \theta)\, \pi(\theta)}{\pi(x \mid \theta, y)\, \pi(y)}$ (Again, RHS may be evaluated at any x for which denominator is non-zero. Useful when full conditional over θ is a GMRF.)

First recall Bayes' rule

$$\pi(\theta \mid y) = \frac{\pi(\theta, y)}{\pi(y)}. \tag{4}$$

Now, we use the product rule to factor the joint distribution $\pi(x, y, \theta)$ in two different ways

$$\pi(x, y, \theta) = \pi(x, y \mid \theta)\, \pi(\theta), \tag{5}$$

and

$$\pi(x, y, \theta) = \pi(x \mid \theta, y)\, \pi(\theta, y). \tag{6}$$

Equating the right-hand sides of (5) and (6) gives

$$\pi(x, y \mid \theta)\, \pi(\theta) = \pi(x \mid \theta, y)\, \pi(\theta, y). \tag{7}$$

Assuming $\pi(x \mid \theta, y) > 0$, we may rearrange (7) to obtain

$$\pi(\theta, y) = \frac{\pi(x, y \mid \theta)\, \pi(\theta)}{\pi(x \mid \theta, y)}. \tag{8}$$

Substituting (8) into (4) gives

$$\pi(\theta \mid y) = \frac{\pi(x, y \mid \theta)\, \pi(\theta)}{\pi(x \mid \theta, y)\, \pi(y)}.$$

as desired.

Question 4.

Assume that $x_1, x_2, \ldots$ are the sequence of states generated by a Markov process (perhaps an MCMC) that is ergodic with respect to some distribution $\pi(x)$. Let $y = g(x)$ for some function g of x, with corresponding pdf $\pi(y)$ (given by the change of variable formula).

(a) Show that $g(x_1), g(x_2), \ldots$ are a the sequence of states generated by a Markov process that is ergodic with respect to $\pi(y)$.

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be the underlying probability space. Let $(X, \mathcal{B}_X)$ be the state space of the stochastic process $(x_n)_{n\geq 1}$. We assume that $(x_n)_{n\geq 1}$ is a *Markov chain*, that is, there exists a transition kernel $P : X \times \mathcal{B}_X \to [0, 1]$ such that for all $n \geq 1$ and $A \in \mathcal{B}_X$,

$$\mathbb{P}(x_{n+1} \in A \mid x_0, \ldots, x_n) = \mathbb{P}(x_{n+1} \in A \mid x_n) = P(x_n, A).$$

We further assume that $(x_n)_{n\geq 1}$ is *ergodic* with respect to a probability measure π on $(X, \mathcal{B}_X)$. Namely, we must have that π is *invariant* for P, i.e., for all $A \in \mathcal{B}_X$,

$$\int_X \pi(dx)\, P(x, A) = \pi(A).$$

And, it satisfies convergence in total variation, i.e, for every starting point $x \in X$,

$$\left\| P^n(x, \cdot) - \pi(\cdot) \right\|_{\mathrm{TV}} \to 0 \qquad (n \to \infty),$$

where P^n is the n-step kernel and $\| \cdot \|_{\mathrm{TV}}$ is total variation distance. Further, let $(Y, \mathcal{B}_Y)$ be a state space, and $g : X \to Y$ a measurable bijection. We define the (clearly) stochastic process $(y_n)_{n\geq 1}$ on $(Y, \mathcal{B}_Y)$ by $y_n := g(x_n)$.

Let us first show that $(y_n)_{n\geq 1}$ is indeed a Markov chain. To this end, let $B \in \mathcal{B}_Y$ and $n \in \mathbb{N}_{\geq 1}$. Then since $(x_n)_{n\geq 1}$ is Markov with transition kernel P,

$$\begin{aligned}
\mathbb{P}(y_{n+1} \in B \mid y_0, \ldots, y_n) &= \mathbb{P}(g(x_{n+1}) \in B \mid x_0, \ldots, x_n) \\
&= \mathbb{P}(x_{n+1} \in g^{-1}(B) \mid x_0, \ldots, x_n)
\end{aligned}$$

$$
\begin{aligned}
&= \mathbb{P}(x_{n+1} \in g^{-1}(B) \mid x_n) \\
&= P(x_n, g^{-1}(B)).
\end{aligned}
$$

But $x_n = g^{-1}(y_n)$ (by bijectivity), hence, $(y_n)_{n\geq 1}$ is Markov with transition kernel

$$Q(y, B) := P\big(g^{-1}(y),\, g^{-1}(B)\big),$$

for all $y \in Y$ and $B \in \mathcal{B}_Y$.

Next, we show that the distribution $\pi_Y : \mathcal{B}_Y \to [0, 1]$ defined by $\pi_Y := \pi \circ g^{-1}$ is invariant for Q. Note that π_Y is indeed a probability measure on $(Y, \mathcal{B}_Y)$ since π is a probability measure on $(X, \mathcal{B}_X)$ and g is measurable. To this end, let $B \in \mathcal{B}_Y$. By definition of Q,

$$\int_Y \pi_Y(dy)\, Q(y, B) = \int_Y \pi_Y(dy)\, P\big(g^{-1}(y),\, g^{-1}(B)\big).$$

Using the definition of π_Y and the change of variable formula, we have

$$\int_Y \pi_Y(dy)\, P\big(g^{-1}(y),\, g^{-1}(B)\big) = \int_X \pi(dx)\, P(x, g^{-1}(B)).$$

By invariance of π for P and the definition of π_Y, we have

$$\int_Y \pi_Y(dy)\, P\big(g^{-1}(y),\, g^{-1}(B)\big) = \int_X \pi(dx)\, P(x, g^{-1}(B)) = \pi(g^{-1}(B)) = \pi_Y(B).$$

Thus π_Y is invariant for Q.

Finally, we show that π_Y satisfies convergence in total variation for Q. Fix $y \in Y$ and let $x = g^{-1}(y)$. Note that one can show via induction that the n-step kernels satisfy

$$Q^n(y, B) = P^n(x, g^{-1}(B))$$

for all $B \in \mathcal{B}_Y$. Therefore, using the definition of total variation, Q, and π_Y, and bijectivity of g,

$$
\begin{aligned}
\big\|Q^n(y, \cdot) - \pi_Y(\cdot)\big\|_{\mathrm{TV}} &= \sup_{B \in \mathcal{B}_Y} |Q^n(y, B) - \pi_Y(B)| \\
&= \sup_{B \in \mathcal{B}_Y} \big|P^n(x, g^{-1}(B)) - \pi(g^{-1}(B))\big| \\
&= \sup_{A \in \mathcal{B}_X} |P^n(x, A) - \pi(A)| \\
&= \big\|P^n(x, \cdot) - \pi(\cdot)\big\|_{\mathrm{TV}}.
\end{aligned}
$$

Since π satisfies convergence in total variation for P, the right-hand side $\to 0$ as $n \to \infty$. Hence π_Y satisfies convergence in total variation for Q as well.

Thus, we have shown that $(y_n)_{n\geq 1}$ is a Markov chain with invariant distribution π_Y and convergence in total variation to π_Y. Hence, $(y_n)_{n\geq 1}$ is ergodic with respect to π_Y, as required.

(b) Show that if the n-step distribution over x_i converges in distribution, then so does the n-step distribution over y_i.

Let us assume that the the n-step distribution over x_i converges in distribution, that is, there exists a random variable x with distribution π such that

$$\mathbb{E}[\varphi(x_n)] \to \mathbb{E}[\varphi(x)] \quad \text{as } n \to \infty,$$

for every bounded, continuous function $\varphi : X \to \mathbb{R}$.

As before, define the random variables $y_n := g(x_n)$ and $y := g(x)$. We further assume that g is continuous and take any bounded, continuous $f : Y \to \mathbb{R}$. Since g is continuous, the composition $f \circ g : X \to \mathbb{R}$ is bounded and continuous. Therefore, since the n-step distribution over x_i converges in distribution to π, we have

$$\mathbb{E}[(f \circ g)(x_n)] \to \mathbb{E}[(f \circ g)(x)] \quad \text{as } n \to \infty.$$

But $(f \circ g)(x_n) = f(g(x_n)) = f(y_n)$ and $(f \circ g)(x) = f(g(x)) = f(y)$, so

$$\mathbb{E}[f(y_n)] \to \mathbb{E}[f(y)] \quad \text{as } n \to \infty.$$

Thus, by definition, the n-step distribution over y_i also converges in distribution as desired.